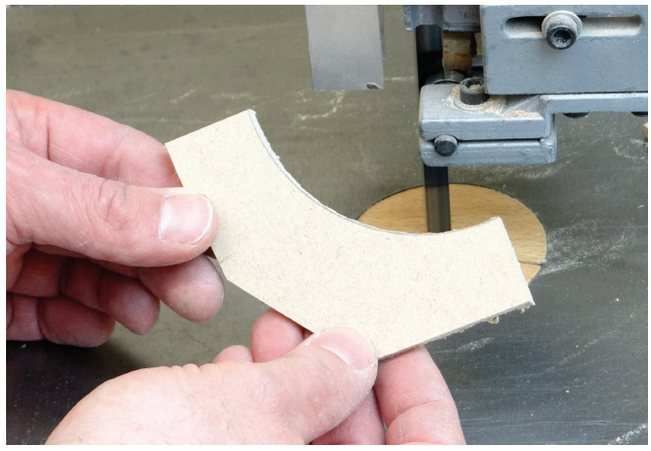
Templates

The next step is to make a template. It is quite possible to make a sphere without a template, but to minimise the work later on, a template will make life much easier. I use an offcut of MDF and draw a curve with a radius to match the sphere, in this case 55mm. The beauty of a sphere is that it should be the same in every direction, so I only make about a quarter of a circle, but you can make as much of the curve as you like.

TOP TIP: PREVENTING BLADE BINDING

Curve cutting on the bandsaw, even with a narrow blade like the 10mm one that I'm using, can be difficult, with the blade binding in the cut. A good way to avoid this is to make a series of relief cuts down to the curve line. This means that there is no material to bind on the blade as you make the cut, allowing you to progress smoothly.

Cutting the template on the bandsaw. Notice the relief cuts which make it much easier to produce a smooth curve

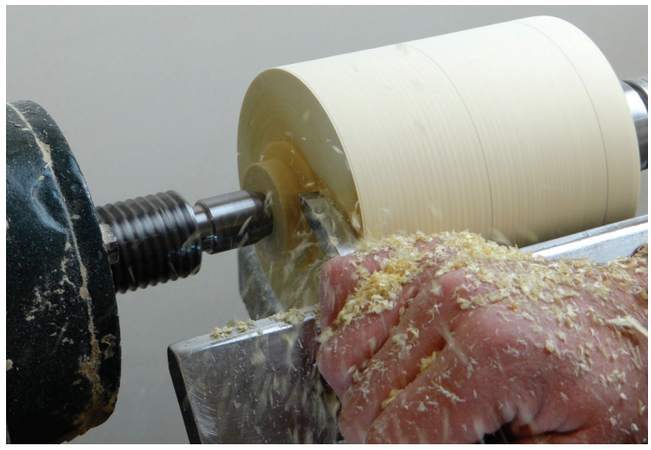


MDF template cut to the radius of the sphere

First stages of turning

With the blank held between centres and roughed to a cylinder of the correct diameter, the next stage is to mark out the sphere. In this case I drew two lines to mark the 110mm sphere and a centre line at 55mm, simply marking them with a pencil and ruler. Using my 10mm beading and parting tool, I then turned down the waste at each end, leaving enough solid timber to give a positive drive, but not so much as to get in the way when shaping the ball.

Marking out the sphere



Using my 12mm spindle gouge I then began to shape the block into a ball. I find it best to do this gradually, first rounding off the corners, then, with each cut, increasing the curve. It is important here to keep an eye on the overall shape; to do this I will tend to work on one side until it begins to resemble a sphere, then go to the other side and balance out the shape. It is important to realise when turning a sphere, that the curve begins on the central line – or the equator if you like – there are no flats on a sphere, just curves, so make sure you cut right from the line.

TERMINOLOGY

For the purposes of clarity, if we imagine the sphere to be the earth, the grain of the timber will be running north to south, from pole to pole – so initially the blank is driven from its poles and the central line is considered the equator. This will help to clarify how the wood is held later on.

Beginning to shape the sphere

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